

Biostatistics — Unit 1

Measures of Variation, Co-variation & Analysis of Variance

PART A: Measures of Variation (Dispersion)

Variation tells us how much the individual observations differ from each other and from the central value. It complements central tendency in fully describing a dataset.

1. Standard Deviation (SD)

$SD = \sqrt{[\Sigma(x_i - \bar{x})^2 / (n-1)]}$ (for sample) $SD = \sqrt{[\Sigma(x_i - \mu)^2 / N]}$ (for population)

Numerical Example:

Diastolic BP (mmHg) of 6 subjects: 70, 75, 80, 85, 90, 95 Mean ($\bar{x}$) = $(70+75+80+85+90+95)/6 = 495/6 = 82.5$ mmHg

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
70	-12.5	156.25
75	-7.5	56.25
80	-2.5	6.25
85	+2.5	6.25
90	+7.5	56.25
95	+12.5	156.25
$\Sigma=495$	$\Sigma=0$	$\Sigma=437.5$

Variance $s^2 = 437.5 / (6-1) = 87.5$ SD = $\sqrt{87.5} = 9.35$ mmHg Normal range = $82.5 \pm 9.35 = [73.15, 91.85]$ mmHg

2. Coefficient of Variation (CV)

$CV = (SD / \text{Mean}) \times 100\%$

CV is dimensionless and used to compare variability across different scales. CV for BP above = $(9.35 / 82.5) \times 100 = 11.33\%$ Rule of thumb: CV < 10% = low variation; 10-30% = moderate; >30% = high variation

3. Standard Error of Mean (SEM)

$SEM = SD / \sqrt{n}$ SEM measures how accurately the sample mean estimates the true population mean.

Using above example: $SD=9.35$, $n=6$ $SEM = 9.35 / \sqrt{6} = 9.35 / 2.449 = 3.82$ mmHg 95% CI for mean = $82.5 \pm 1.96 \times 3.82 = [75.0, 90.0]$ mmHg

PART B: Co-variation — Correlation Analysis

Co-variation (correlation) measures the degree and direction of relationship between two variables.

1. Pearson's Correlation Coefficient (r)

$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{\sum(x_i - \bar{x})^2 \times \sum(y_i - \bar{y})^2}}$ Range: $-1 \leq r \leq +1$
 $r = +1$: Perfect positive correlation
 $r = 0$: No correlation
 $r = -1$: Perfect negative correlation

Numerical Example:

Age (x) and Systolic BP (y) of 5 patients:

Patient	Age (x)	SBP (y)	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
1	25	110	-15	-18	+270	225	324
2	35	120	-5	-8	+40	25	64
3	40	128	0	0	0	0	0
4	50	138	+10	+10	+100	100	100
5	60	150	+20	+22	+440	400	484
Mean	$\bar{x}=42$	$\bar{y}=129.2$			$\Sigma=850$	$\Sigma=750$	$\Sigma=972$

$r = 850 / \sqrt{750 \times 972} = 850 / \sqrt{729000} = 850 / 853.8 = 0.996$
Interpretation: Very strong positive correlation between age and BP ($r \approx +1$).

2. Spearman's Rank Correlation (rs)

Used for ordinal data or when Pearson assumptions are violated. $r_s = 1 - [6 \times \sum d^2] / [n(n^2 - 1)]$ where d = difference in ranks

Example:

Subject	Rank in Test A	Rank in Test B	d = RA-RB	d ²
1	1	2	-1	1
2	2	1	+1	1
3	3	4	-1	1
4	4	3	+1	1
5	5	5	0	0
			Σd^2	4

$$r_s = 1 - [6 \times 4] / [5 \times (25 - 1)] = 1 - 24/120 = 1 - 0.2 = 0.8$$
 Strong positive rank correlation.

PART C: Analysis of Variance (ANOVA)

ANOVA tests whether means of three or more groups are significantly different. It partitions total variation into 'between-group' and 'within-group' components.

Key Concept: Partitioning of Variance

Total SS = Between-Group SS + Within-Group SS $SST = SSB + SSW$ F-ratio = MSB / MSW where $MSB = SSB/(k-1)$ and $MSW = SSW/(N-k)$ k = number of groups, N = total observations

Detailed One-Way ANOVA Example:

Three diets tested on 3 groups (n=4 each). Weight loss (kg) after 8 weeks:

Group A (Diet 1)	Group B (Diet 2)	Group C (Diet 3)
3	5	8
4	6	9
5	7	7
4	6	8
$\Sigma A=16, \bar{x}_A=4$	$\Sigma B=24, \bar{x}_B=6$	$\Sigma C=32, \bar{x}_C=8$

Grand Mean = $(16+24+32)/12 = 72/12 = 6.0$ kg SSB (between) = $n \times \Sigma(\bar{x}_j - \text{Grand Mean})^2 = 4 \times [(4-6)^2 + (6-6)^2 + (8-6)^2] = 4 \times [4 + 0 + 4] = 32$ SSW (within) = $\Sigma(\text{each value} - \text{its group mean})^2$ Group A: $(3-4)^2 + (4-4)^2 + (5-4)^2 + (4-4)^2 = 1+0+1+0 = 2$ Group B: $(5-6)^2 + (6-6)^2 + (7-6)^2 + (6-6)^2 = 1+0+1+0 = 2$ Group C: $(8-8)^2 + (9-8)^2 + (7-8)^2 + (8-8)^2 = 0+1+1+0 = 2$ $SSW = 2+2+2 = 6$ $MSB = SSB/(k-1) = 32/2 = 16$ $MSW = SSW/(N-k) = 6/9 = 0.667$ $F = MSB/MSW = 16/0.667 = 24.0$

Critical F at $\alpha=0.05$ with $df_1=2$, $df_2=9$ is approximately 4.26. Calculated $F = 24.0 >> 4.26 \rightarrow$ Reject H_0 Conclusion: The three diets produce significantly different weight loss.

ANOVA Summary Table

Source	SS	df	MS	F
Between Groups	32	2	16.0	24.0
Within Groups (Error)	6	9	0.667	—
Total	38	11	—	—